

Morris Greenberg

Ph.D. Candidate in Statistical Sciences

📍 Toronto, ON, Canada

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🌐 morrisgreenberg

Education

PhD	Department of Statistical Sciences, University of Toronto , Statistical Sciences • Advisors: Prof. Radu Craiu, Prof. Kieran Campbell • Research interests: Bayesian modeling and programming, scalable methods for complex health-care data (especially ‘omics data), graph inference, hierarchical models, approximate Bayesian computation	Toronto, ON, Canada Sept 2021 – present
MS	Department of Statistical Science, Duke University , Statistical Science	Durham, NC, USA Aug 2019 – May 2021
BS	Tufts University , Quantitative Economics and Mathematics	Medford, MA, USA Aug 2012 – May 2016

Awards

NSERC Postgraduate Scholarship - Doctoral \$101,000 CAD	Canada Sept 2023
Queen Elizabeth II Graduate Scholarship in Science and Technology \$15,000 CAD	Ontario Sept 2022
Doctoral Recruitment Award \$5,000 CAD	Toronto, ON Sept 2021
BEST Award for Master’s Research in Statistical Science \$500 USD	Durham, NC June 2021

Conference Presentations

- Greenberg, M., Campbell, K., Craiu, R. (2026, June). Estimating Hierarchical Tissue Organization with Structure Learning. 2026 International Society for Bayesian Analysis World Meeting, Nagoya, Japan. (Poster)
- Greenberg, M., Campbell, K., Craiu, R. (2025, September). Restricted Search Space Graph MCMC via Birth-Death Processes. The Fast and Curious 2: MCMC in Action, Toronto, ON, Canada.
- Greenberg, M., Campbell, K., Craiu, R. (2025, May). Restricted Search Space Graph MCMC via Birth-Death Processes. Statistical Society of Canada Annual Meeting 2025, Saskatoon, SK, Canada. Advances in Statistical Inference and Computing.
- Greenberg, M., Campbell, K., Craiu, R. (2025, January). Restricted Search Space Graph MCMC via Birth-Death Processes. University of Florida Department of Statistics' Annual Winter Workshop: 2025 Computational Methods in Bayesian Statistics, Gainesville, FL, USA. (Poster)
- Greenberg, M., Craiu, R., Campbell, K. (2024, June). Restricted Search Space Graph MCMC via Birth-Death Processes. International Chinese Statistics Association Canada Chapter Symposium, Niagara Falls, ON, Canada. Advances in Monte Carlo Methods.
- Greenberg, M., Craiu, R., Campbell, K. (2023, May). Restricted Search Space MCMC With Adaptive Weighting And Sparsity Parameterization For Graph Inference. Statistics Society of Canada Meeting 2023, Ottawa, ON, Canada. Advances In Bayesian Modelling And Computation.
- Greenberg, M., Craiu, R., Campbell, K. (2023, May). Restricted Search Space MCMC Methods for Graph Inference. The Fast and Curious: Modern Markov Chain Monte Carlo, Minneapolis, MN, USA.
- Greenberg, M., Ma, L., Wang, Z., Ma, P., Sung, A. (2021, August). Logistic Tree Gaussian Processes (LoTGaP) for the Microbiome. Joint Statistical Meetings 2021, Next-Generation Sequencing and High-Dimensional Data Section.

Publications

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| Restricted Search Space Graph MCMC via Birth-Death Processes | 2026 |
| Morris Greenberg, Kieran R. Campbell, Radu V. Craiu
arxiv.org/abs/2604.10863 (arXiv:2604.10863 [stat]) | |
| Chlorhexidine Gluconate Bathing Reduces the Incidence of Bloodstream Infections in Adults Undergoing Inpatient Hematopoietic Cell Transplantation | Jan 2021 |
| Vinay K. Giri, et al.
10.1016/j.jtct.2021.01.004 (Transplantation and Cellular Therapy) | |
| Development of predictive risk models for major adverse cardiovascular events among patients with type 2 diabetes mellitus using health insurance claims data | 2018 |
| J.B. Young, M. Gauthier-Loiselle, R.A. Bailey, A.M. Manceur, P. Lefebvre, Morris Greenberg, M.H. Lafeuille, M.S. Duh, B. Bookhart, C.H. Wysham
(Cardiovascular Diabetology, 17(1), p.118) | |
| Development of Risk Models for Major Adverse Chronic Renal Outcomes Among Patients with Type 2 Diabetes Mellitus Using Insurance Claims - A Retrospective Observational Study | 2019 |
| C.H. Wysham, M. Gauthier-Loiselle, R.A. Bailey, A.M. Manceur, P. Lefebvre, Morris Greenberg, M.S. Duh, J.B. Young
(Current Medical Research and Opinion) | |

Experience

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| University of Toronto Statistical Sciences Department , Course Instructor | Toronto, ON |
| • Course co-instructor for An Introduction to Statistical Reasoning and Data Science (STA 130) | Jan 2025 – May 2025
5 months |

University of Toronto Statistical Sciences Department, Teaching Assistant

- Course list includes: Advanced Computational Methods for Statistics I (STA 2311, Fall 2024), Bayesian Statistics (STA 365, Winters 2022-2025, Fall 2025), Statistical Consultation, Communication, and Collaboration (STA 490, 2022-2025), Introduction to Machine Learning (STA 314, Fall 2022, Winter 2026), Data Science in Practice (STA 2546, Winter 2022)

Toronto, ON
Aug 2021 – present
5 years

Li Ma Statistics Lab / Duke University Hospital, Research Assistant

- Developed novel hierarchical models using Gaussian processes to analyze the dynamics of microbiome composition of patients in bone marrow transplant studies
- Designed a microbiome logistic tree package in R that includes microbiome methods for mixed effects modeling, Gaussian process modeling, and topic modeling

Durham, NC
May 2020 – Aug 2021
1 year 4 months

Analysis Group, Analyst / Senior Analyst

- Developed models to predict cardiovascular risk, chronic kidney disease, and chronic thromboembolic pulmonary hypertension (CTEPH) in Type II Diabetes Mellitus patients, which resulted in multiple academic journal articles
- Created an elasticity model to describe the price sensitivity of patients, physicians, and providers towards cholesterol lowering drugs in expert reports for intellectual property litigation
- Built machine learning tools (e.g. neural networks, random forests, Gaussian mixture models) to identify user types of opioids for the purposes of targeted treatment care
- Analyzed the source code for millions of files across programming languages for tax litigation; measured the similarity between pairs of files to understand how source code was modified by developers over time

Boston, MA
Aug 2016 – May 2019
2 years 10 months

Washington Nationals, Baseball Operations Intern / Consultant

- Derived, coded, and maintained a Markov Chain simulator for the team
- Worked on analytical problems using statistical learning methods (e.g. random forests, gradient boosted algorithms, and generalized linear models) in R

Washington, DC
May 2015 – Apr 2016
1 year

Technical Skills

Programming: R, Python, C++, STAN, Matlab, LaTeX, Stata, SAS, SQL (MySQL, Oracle SQL Developer)

Languages: English, Spanish, Hebrew

Miscellaneous: Top 100 Scrabble Player in North America; came in 10th at the 2017 North American Championships